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Course: Business Analytics - Foundations

The Language of Business Analytics

Research Assignment: Module 1

1. Foundation Concepts of Analytics

1. Business Analytics (BA)

Definition: Business analytics is using data and math to solve business problems and make better business decisions. Its focused on company goals like profit, growth, or efficiency. The main aim of business analytics is to use the data to drive decisions into strategic outcomes instead of just reporting. (Yin a Fernandez, 2020)

Example: A retailer like Shoprite uses business analytics to figure out which store locations should stock more cold drinks before a heat wave, to boost sales.

2. Data Analytics (DA)

Definition: Data analytics is the broader process of examining raw data to find patterns, trends and answers it does not have to be for business; scientists or sports teams use it too. (Sarker, 2021)

Distinction: it is a toolkit that includes descriptive and advanced analytics, Business Analytics is how a company applies the toolkit to make decisions. (Sarker 2021).

Example: Woolworths' data team analyse customer wait times to find that Friday at 12pm are always

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worse, so they add more staff then.

3. Business Intelligence (BI)

Definition: Business Intelligence is the practice of turning historical data into accessible information through reporting and visualisation, to understand what happened and why. Visualisation is the use of dashboards, charts or reports; emphasise ~~of~~ on descriptive analytics and operational visibility but not on what will happen in the future. (Sun et al., 2015)

Distinction: BI primarily provides visibility and insight from past data, while BA and Data Analytics extend into future planning and optimisation driven analytics. (Sun et al., 2015)

Example: Takealot's manager opens a BI dashboard every morning to see yesterday's sales by product and region.

4. Data Driven Decision Making

Definition: The practice of using facts and numbers, not just gut feel, to choose what to do next. It pulls together information about what happened in the past, what might happen next and what should be done to guide the decisions and rules (Araz et al., 2020). Insights are from descriptive, predictive and prescriptive analytics to guide actions and governance (Araz et al., 2020).

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Distinction: BI mainly just shows reports about the past, while Data-driven decision making goes a step further by using the data to decide. (Aroz et al., 2020.)

Example: A logistics company tracks fuel use per route of its trucks. Data shows Route B uses 20% more diesel than Route A, so they switch trucks to Route A.

5. Big Data

Definition: Big Data is datasets that are large, diverse and characterised by volume, variety and speed (velocity). They cannot be handled by normal Excel. They need special tools like Hadoop or cloud computing (Islam, 2024)

Example: Vodacom analyses billions of call records and location pings daily to spot network outages and drop call patterns in real time.

6. Descriptive Analytics

Definition: Descriptive analytics answers "what happened?". It summarises past data into totals, averages and charts, basically the foundation of analysis. One cannot diagnose or predict without first describing. (Basu et al., 2023).

Example: A bakery runs a report with the following findings: Sold 1200 loaves of bread sold in March,

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compared to 1000 loaves in February.

7. Diagnostic Analytics

Definition : It answers the question "why did it happen?". It digs deeper into descriptive data to find causes. Investigations are done by using drill-downs, correlations or root-cause analysis (Basu et al., 2023.)

Example : If sales went up by 20% , diagnostic Analytics shows the reasons why ~~that~~ sales went up or what made them lower the previous month. We would look at logistic constraints that led to delayed delivery in the ~~perfor~~ previous month or price reductions in the current month.

8. Predictive Analytics

Definition : It answers the question "what will likely happen next?" It uses stats, machine learning and past patterns to forecast, giving probabilities. It gives foresight for planning and risk management. (Basu et al., 2023.)

Distinction : It focuses on forecasting future states, while descriptive and diagnostic ~~expl~~ explain past states ; prescriptive analytics then recommends actions based on predictions (Basu et al., 2023.)

Example : A medical aid company uses predictive models to estimate which clients are most likely to claim in the next year, so that they can

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price their policies better.

9. Prescriptive Analytics

Definition: It answers the question 'what should we do about it?' Prescriptive Analytics, extends predictions to recommend specific actions and quantifies the expected impact. (Sheng et al., 2020)

Example: Eskom's grid software predicts a demand spike between 6-9 pm and prescribes which power stations to ramp up to avoid loadshedding.

10. Data Literacy

Definition: It is the ability to read, understand, question and argue with data. It is about knowing what a chart means, spotting bad statistics and asking relevant question on the data. (Yin & Fernandez, 2020)

Example: A warehouse supervisor looks at a chart and notices the scales is misleading. Because he is data literate, he asks for raw numbers for comparison before approving overtime.

2: Data, Measurement & Reporting

11. Structured Data

Definition: Structured data is information that fits neatly into rows and columns, like a spreadsheet or database table. It has a clear format so computers

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can easily search, sort and analyse it. Each field has a defined type such as date, number or text. (Saker, 2021; Yin & Fernandez, 2020)

Example: A bank's customer table with columns for Account number, name, balance, branch and the date the account was opened.

12 Unstructured Data

Definition: Unstructured data has no predefined format or schema (eg. videos, social media posts and voice recordings). It makes it harder for traditional processing and requires more complex methods. (Saker, 2021)

Example: MTN analysing thousands of customer call center audio recordings to detect common complaints.

13. Data Warehouse.

Definition: It is a central system that stores structured data from multiple sources, cleaned and organised for reporting and analysis. (Yin & Fernandez, 2020)

Distinction: A data warehouse emphasises historical data for reporting, while data lakes may store both structured and unstructured data with more flexible schemas.

Example: Woolworths pulls sales inventory and

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supplier data into a data warehouse so that managers can run fast reports across all stores

14. Data Mining

Definition: It is the process of searching large datasets to discover hidden patterns, correlations or rules that aren't obvious. It uses machine learning and statistics to find insights that can predict behaviour or improve decisions. (Saker, 2021; Yin & Fernandez, 2020.)

Example: Discovery Health mines claims data and finds that members who buy running shoes are 30% less likely to claim for heart issues.

15. Data Visualisation

Definition: Data visualisation is turning numbers and data into charts, graphs or maps so that people can understand the data quickly. Its main goal is to make ~~me~~ complex data easy to see and interpret. A good visual highlights trends or outliers without the need for a spreadsheet. (Nraiji et al., 2024; Yin & Fernandez, 2020.)

Example: Uber Eats uses a heatmap visualisation to show restaurant owners which suburbs order the most at 7-10pm on Fridays.

16. Dashboards

Definition: It is a single screen that pulls together key charts, metrics and KPI's so a

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des decision maker can monitor performance at a glance. It updates automatically and is interactive, unlike a static report. (Basu et al., 2023; Nnaji et al., 2024)

Example 3

A standard Bank branch manager uses a dashboard to track daily loan approvals, ATM uptime and customer waiting times in real time.

11. Metric

Definition: A metric is any quantifiable measure used to track and assess some aspects of performance. It may be descriptive or diagnostic but is not tied to performance objectives (Saker, 2021; Yin & Fernandez, 2020)

Distinction: All KPI's are metrics, but not all metrics qualify to be KPI's. KPI's are metrics that are strategically linked to goals and constrained by business targets. (Nnaji et al., 2024; Yin & Fernandez, 2020),

Example: The number of website visits is a metric for most companies. It becomes a KPI if the company depends on the traffic, then they would need to track progress towards a specified traffic goal eg 10000 visits per week. (Nnaji et al., 2024; Yin & Fernandez, 2020)

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18. Key Performance Indicator (KPI)

Definition: A KPI is a specific metric that is directly tied to a critical business goal. It measures how well the company is doing at something that matters to ~~se~~ success. If a metric does not influence strategy or target then it is not a KPI. (Nnaji et al., 2024; Yin & Fernandez, 2020).

Distinction: Every KPI is a metric, but a metric only becomes a KPI when it is strategically meaningful with a clear goal and monitoring plan (Nnaji et al., 2024; Yin & Fernandez, 2020)

Example: For an airline like Fly Safair, "~~On-time~~ 'On-time departure rate' is a KPI because it drives customer satisfaction and repeat bookings

19. Benchmark

Definition: A benchmark is a standard or point of reference used to compare performance against others or against past results. It answers "are we good compared to X?" Benchmarks can be internal, competitor based or industry wide. (Nnaji et al., 2024; Yin & Fernandez, 2020).

Distinction: Benchmarks provide external and internal context for performance, they are not a metric or KPI, but a basis for setting targets and evaluating relative performance. (Nnaji et al., 2024; Yin & Fernandez, 2020.)

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Example : A car dealership in Alberton benchmarks the number of days it takes to sell a specific car model against the national average of 45 day. This is to check if their sales team is under performing.

20: Segmentation

Definition : Segmentation is splitting a broad group of customers, products or dealer into smaller groups that share similar traits. This lets a business tailor strategies (marketing, production or service) instead instead of using a one-size fits all approach. Segments are based on behavior, demographics or value. (Basu et al., 2023)

Distinction : Segmentation is an analytical approach used to interpret data within groups; it is not a single metric but a method that informs KPI definition and targeting strategies. (Basu et al., 2023)

Example : Pict n Pay segments loyalty card users into "budget families" versus "heart-focused singles" so that it can send different promotions to each group.

3: Customer & Sales KPIs

21. Conversion Rate

Definition : Conversion rate measures the percentage of people who take a desired action out of everyone

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who had the chance. The action could be a purchase, sign up or download depending on the goal. It answers "how effective are we at turning interest into results?" (Basu et al., 2023)

Example: A Mr D Food restaurant ad gets 1000 views and 50 orders. The conversion rate is 5%, showing how well the ad turned viewers into customers.

22. Customer Acquisition Cost (CAC)

Definition: CAC is the total money a business spends on sales and marketing to gain one new customer. It includes ad spend, staff salaries and tools divided by the number of new customers in that period. If CAC is too high, growth becomes unprofitable. (Basu et al., 2023)

Distinction: CAC is a metric, it becomes a KPI when linked to profitability targets and compared against Customer Lifetime Value (CLV) to assess efficiency of growth. (Basu et al., 2023.)

Example: A fintech start up in Guangzhou spends R200 000 on ads and gets 400 new users. $CAC = R500$ per customer.

23. Customer Lifetime Value (CLV/LTV)

Definition: CLV is the total net profit a company expects to earn from one customer over the entire relationship. It factors in how often they

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buy, how long they stay. It helps decide how much you can afford to spend to acquire them. (Basu et al., 2023).

Distinction: CLV is a metric; it becomes a KPI when benchmarked against CAC or targeted to improve gross margins over time. The CLV:CAC ratio is a widely watched health metric (Basu et al., 2023).

Example: A gym has members who stay 24 months on average and pay R400 per month. CLV is R9600 minus costs, so they know not to spend more than that to acquire a member.

The Most Important Ratio CLV:CAC

- This ratio shows if a customer is worth more than it costs to get them on board. A healthy ratio is usually 3:1 or higher, meaning you earn R3 for every R1 spent on acquisition. If the ratio is 1:1 or less, the business loses money on each customer and can't survive long term, even with fast growth (Basu et al., 2023).

24 Churn Rate

Definition: Churn rate is the percentage of customers who stop using a product or service in a given period. It answers "how many people are we losing?" High churn kills growth because a company is constantly replacing lost customers (Basu et al., 2023).

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Example: Netflix South Africa starts January with 100 000 subscribers and loses 4000 by month end. Churn Rate = 4% for the month.

25. Customer Retention Rate.

Definition: It measures the percentage of customers a business keeps over a period, reflecting loyalty and satisfaction. Retention is the opposite of churn. (Basu et al., 2023.)

Example: An online store had 500 customers last year and gained 200 new ones. At the end of the new year they have a total of 600 clients. Retention rate is $(600 - 200) / 500 = 80\%$

26 Net Promoter Score (NPS)

Definition: NPS measures customer loyalty by asking "How likely are you to recommend us to a friend?" It reflects loyalty and satisfaction and indicates how healthy a brand is. Scores are normally between 0-10; where Promoters 9-10, Passive 7-8, and Detractors 0-6. $NPS = \% \text{ Promoters} - \% \text{ Detractors}$. (Basu et al., 2023)

Example: After purchasing RFC dined wings, 70% of the clients are promoters and 10% are detractors, $NPS = 60\%$ which is considered excellent

27. Average Order Value (AOV)

Definition: AOV is the average amount of money

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a customer spends each time they place an order. It is calculated as $\text{total revenue} \div \text{number of orders}$. Raising AOV is one of the fastest ways to grow revenue without finding new customers. (Basu et al., 2023)

Example: Takcalot made £1 000 000 from 20 000 orders last week. $\text{AOV} = £1\,000\,000 / 200\,000 = £50$.

They may then test 'free shipping over £75 orders' to push AOV higher.

28 sales Funnel

Definition: A sales funnel is the step-by-step journey, from first awareness to final purchase of a potential customer. Stages usually include Awareness, Interest, Consideration and Action. It helps businesses see where prospects drop off. (Basu et al., 2023)

Example: A car dealership's funnel: 10 000 people see an advertisement, 1000 of them visit the website, 100 of them book a test drive and 20 of those purchase the car. Marketing fixes the leak between test drive and purchase.

4: Marketing & Campaign Analytics

29. Marketing Campaign

Definition: A marketing campaign is a coordinated set of activities designed to promote a product, service or brand to a specific audience over a set period. It uses

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channels like email, social media or TV with a clear goal such as sales, signups or awareness. Campaigns are planned, executed and analysed using data at different touchpoints. (Basu et al., 2023)

Example: Castle Lite runs a "Summer Refresh" campaign in December using billboards, Instagram reels and radio ads to drive beer sales in Gauteng. Campaign effectiveness will be evaluated after the period using post-campaign analysis, ROI metrics and attribution to determine its effectiveness. (Basu et al., 2023)

30. Post Campaign Analysis

Definition: Post-campaign analysis is the review you do after a campaign ends, used to evaluate what worked and what did not. Results are compared to goals using metrics such as incremental revenue, ROI and ~~attributed~~ attribution to determine which elements contributed the most. This will determine whether the different campaign aspects should be repeated, fixed or whether the whole campaign should be cancelled next time. (Basu et al., 2023)

Example: After Black Friday, Takealot's team does post-campaign analysis and finds email drove 40% of sales but TikTok ads had negative Return on Ad Spend (ROAS). Next Black Friday they may reallocate budget to another medium that may bring in more revenue (Basu et al., 2023)

31. Incremental Revenue

Definition: Incremental revenue is the extra sales a campaign

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or marketing activity generated that would not have happened otherwise. It excludes baseline sales to measure true campaign impact, as total revenue alone can be misleading. (Basu et al., 2023).

Example: Nando's runs a TV ad and total sales rise by R2 million. But data shows R1.2 million would have happened anyway. Incremental revenue = R800 000; which is $R2000\ 000 - R1200\ 000$ (Basu et al., 2023)

32 Return on Investment (ROI)

Definition: ROI measures overall profit from an investment, relative to its cost, as a percentage. For campaigns, it is $(\text{Revenue from campaign} - \text{Cost}) / \text{cost} \times 100$, it answers "was this worth it?" across all costs, not just ad spend. Post-campaign analysis always calculates ROI. (Basu et al., 2023)

Example: Capitec spends R500k on a campaign and makes R2 million in incremental revenue with R300k in product costs. $ROI = (R2m - R300k - R500k) / R500k \times 100$
 $= 240\%$

33. Return on Ad Spend (ROAS)

Definition: ROAS is similar to ROI, but only looks at ad spend and not total cost. Formula: $\text{Revenue from ads} / \text{Ad spend}$. It is mostly used to judge channels like Google and Facebook. A campaign can have great ROAS but bad ROI if other costs are high. (Basu et al., 2023).

Example: A Shein campaign spends R100k on Instagram

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ads and generates R400k in tracked sales, ROAS = 4:1 or 400%.

34. Click-Through Rate (CTR)

Definition: CTR is the percentage of people who clicked an ad or link after seeing an online ad, relative to the number of times it is shown. Formula: $\text{Clicks} / \text{Impressions} \times 100$. It measures how compelling the creative or message is. Low CTR often triggers A/B (split testing) to find better headlines or images. (Basu et al., 2023)

Example: A superbasket banner ad gets 200k impressions and 2000 clicks. $\text{CTR} = 1\%$, which the company compares to the 0.5 - 2% industry benchmark.

35. Bounce Rate

Definition: Bounce rate is the percentage of visitors who land on a webpage and leave without taking any action or visiting another page. A high bounce rate suggests the page did not match the ad or loaded slowly. It's a key signal in post campaign analysis of landing pages. (Basu et al., 2023)

Example: Vodacom's iPhone promo ad has 2% CTR, but the landing page bounce rate is 85%. Marketing Attribution says the ad worked, but the page did not bring conversions.

36. A/B Testing

Definition: A/B testing compares two versions of an ad, email or webpage to see which performs better on a

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metric like CTR or conversion rate. Version A is shown to half on audience and B to the other half. (Basu et al., 2023)

Example: Travelstart A/B tests two email subject lines "Deals to Cape Town" vs "Your Cape Town Escape Awaits". Version B gets 30% more clicks, so they use it for all clients (full send).

31. Lead Generation

Definition: Lead generation is the process of attracting and capturing interest from potential customers who might buy later, forming the top of the sales funnel. A lead is someone who gave their contact information (phone/email). Campaigns often optimise for leads instead of direct sales, especially in the car and insurance space. (Basu et al., 2023)

Example: Outsurance run a Facebook campaign offering a free car insurance quote. Each form completed is a lead for their sales team.

38. Marketing Attribution

Definition: Marketing attribution is figuring out which channels or touchpoints deserve credit for a sales or lead. It is critical for calculating true ROAs and doing post campaign analysis. (Basu et al., 2023)

Example: A customer sees a Mr Price YouTube ad, clicks a Google search ad a week later, then buys via email. Last-click attribution gives 100%

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credit to email, but multi-touch attribution splits it across all three channels.

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